EXPERIMENT-06:

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

CODE:

```
# Import required libraries
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

# Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create and train the Naive Bayes model
model = GaussianNB()
```

```
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")

# Display confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Plot confusion matrix
plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)
```

```
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Naive Bayes Confusion Matrix")
plt.show()

# Display classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
```

OUTPUT:

```
Accuracy: 97.77777777777777 %

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

